

ANDILE MHLANGA

Full Stack Software Engineer
React · TypeScript · Node.js · PostgreSQL

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PROFESSIONAL SUMMARY

Full Stack Software Engineer based in Johannesburg, South Africa, with hands-on experience building production-grade web applications using React, TypeScript, Node.js, and PostgreSQL. Skilled in designing scalable REST APIs, implementing role-based access control, integrating AI services, and deploying on cloud platforms including Railway and Vercel. Brings a rare combination of real-world operational experience in control room and security environments and the technical ability to build systems that solve those exact problems. Currently expanding into C# and the .NET framework. Passionate about clean architecture, real-world impact, and shipping software that works.

AREA OF EXPERTISES

- **Frontend Development:** React · TypeScript · HTML5 · CSS3 · Responsive Web Design
- **Backend Development:** Node.js · Express.js · REST APIs · JWT Authentication · RBAC
- **Database & ORM:** PostgreSQL · MongoDB · SQL Server · Prisma · Supabase · CRUD Operations
- **AI & Integrations:** OpenAI API · AI-generated reporting pipelines · Prompt engineering
- **Tools & Infrastructure:** Git · GitHub · Docker · GitHub Actions · Railway · Vercel · CI/CD

PROJECTS

FLAWS - Full-Stack eCommerce Store | *React · TypeScript · Node.js · PostgreSQL · Prisma · Supabase · Vercel* www.flawswrldwide.com

- Built and deployed a fully production-ready eCommerce platform selling apparel to customers across South Africa - live at flawswrldwide.com.
- Developed a complete Admin Panel with inventory control, product/collection management, order oversight, waitlist activation, and an activity log for full operational visibility.
- Integrated automated order fulfillment with The Courier Guy API, enabling real-time shipping dispatch and tracking without manual intervention.
- Implemented a waitlist system allowing early-access signups before launch, and built a full authentication flow with role-based access separating admin and customer experiences.
- Managed the full product database via Prisma ORM on Supabase-hosted PostgreSQL, with image storage, stock tracking, and low-stock alerting built into the admin dashboard.

SecurePulse | *React · TypeScript · Node.js · Express · PostgreSQL · OpenAI API · JWT/RBAC · Railway* github.com/andile593/SecurePulse

- Built a full-stack security operations platform that replaces traditional control rooms with a centralised, AI-assisted digital workspace for security companies managing multiple client sites.
- Implemented AI-generated incident reporting: guards complete a structured form and the platform automatically produces a formatted, professional incident report via the OpenAI API - eliminating manual write-ups.
- Designed a role-based access control system with four distinct roles (Administrator, Control Room Operator, Field Guard, Site Manager), each with scoped permissions and JWT-authenticated sessions.
- Architected a clean REST API with separate services for alarm monitoring, incident logging, AI report generation, and client/site management - deployed on Railway with 107 commits.

Zimba-Moves | *React · TypeScript · Node.js · Express · PostgreSQL · Prisma · JWT/RBAC* github.com/andile593/Zimba-Moves

- Developed a full-stack marketplace platform connecting customers with verified moving companies across South Africa - handling the entire workflow from quote request to booking confirmation and post-move reviews.

- Implemented a competitive quoting system allowing multiple moving companies to submit quotes on a single request, enabling customers to compare and accept offers within the platform.
- Built an admin verification layer requiring moving companies to pass an approval process before listing on the platform, ensuring only legitimate operators participate.
- Designed multi-role architecture (Customer, Moving Company, Administrator) with isolated permissions and a clean PostgreSQL schema managed via Prisma migrations.

TechCare - Multi-Tenant Healthcare Backend API | Node.js · TypeScript · Express · PostgreSQL · Prisma · Docker · GitHub Actions github.com/andile593/TechCare

- Architected a multi-tenant backend from a relational schema up, enforcing organization-level data isolation across patients and clinical records to prevent cross-tenant data exposure.
- Built a secure authentication system with httpOnly-cookie JWT sessions and refresh-token rotation, including a reuse-detection design intended to revoke a compromised session family on replay of a dead token.
- Enforced data integrity at the database level - foreign-key constraints, cascade rules, and soft deletes - rather than relying solely on application-layer checks.
- Set up a CI/CD pipeline in GitHub Actions - lint, type-check, and test against an ephemeral PostgreSQL instance on every PR, with branch protection gating merges and automated Docker image builds to GHCR on release.
- Enforced strict TypeScript (exactOptionalPropertyTypes, noUncheckedIndexedAccess) to catch data-handling bugs at compile time.

WhatsApp-Shift-Reporting-Tool | Python · Automation · WhatsApp Web github.com/andile593/WhatsApp-Shift-Reporting-Tool

- Built a Python automation system that captures screenshots from an operational monitoring system and sends them to a designated WhatsApp group at scheduled intervals - born directly from control room experience.

WORK EXPERIENCE

Control Room Operator | [Hooray Power - Bedfordview](#)

Aug 2025 – Present

- Monitor solar, battery, and power systems in real time; investigate alerts and respond to operational events.
- Report outages, system faults, and anomalies; escalate to relevant engineering teams when needed.
- Respond to client and staff communications; coordinate incident resolution end-to-end.
- Perform routine system health checks and document performance metrics and incidents.
- Follow SOPs and emergency procedures to ensure safe and reliable system operation.

Control Room Operator | [Silver Security Group - Heidelberg](#)

May 2025 – Aug 2025

- Monitored CCTV and alarm systems providing 24/7 surveillance across multiple client sites.
- Tracked armed response vehicles in real time via Cartrack telematics, ensuring patrol compliance and rapid deployment.
- Logged vehicle entries, patrols, and incidents in the Occurrence Book; completed detailed shift handovers.
- Responded to alarms, coordinated with field guards, and escalated issues as required.

EDUCATION

National Senior Certificate (Matric)

2023

Nigel Secondary School · Nigel, Gauteng

CERTIFICATIONS & LICENSES

- PSIRA Grade EDC - April 2025
- Code 10 Driver's License (C1)